

- 10 (a)** e.g. zero output resistance / impedance
infinite bandwidth
infinite slew rate
1 mark each, max. 3

B3 [3]

- (b) (i)** at 1.0°C , thermistor resistance is $3.7\text{ k}\Omega$
amplifier gain $= -R/740 = -3700/740$ (*negative sign essential*)
 $= -5.0$

B1
C1
C1

$$\text{potential} = 1.0 / -5.0 = -0.20\text{ V}$$

A1 [4]

- (ii)** at 15°C , $R = 2.15\text{ k}\Omega$ (*allow $\pm 0.05\text{ k}\Omega$*)
reading $= (2150/740) \times 0.2$
 $= 0.58\text{ V}$ ($0.59\text{ V} \rightarrow 0.57\text{ V}$)

C1
A1 [2]

- (c) (i)** 0.68 V

A1 [1]

- (ii)** resistance (of thermistor) does not change linearly with temperature

B1 [1]